

Comprehensive Academic Report: Digital Credit Engineering and the Apyx Financial Protocol

An in-depth study on converting institutional cash flows into sustainable decentralized yields

1. Economic Background and the "Digital Credit" Thesis

The Yield Crisis in the Stablecoin Market

- Stablecoins represent the largest opportunity in all of crypto, yet they are fundamentally broken because they offer virtually no native yield.
- The structural economic dilemma is that holding traditional fiat-backed stablecoins, such as USDT or USDC, does not generate an automatic yield for users.
- While traditional stablecoins lose their purchasing power through inflation, the issuers retain the yield generated by the reserves rather than passing it on to holders.

What is Digital Credit and the Flywheel Effect?

- Digital Credit is an innovative asset class that brings dividend-backed yield powered by publicly traded preferred equity to onchain finance.
- This integration creates a "Digital Credit Flywheel," where the issuance of preferred shares generates monthly USD dividend cash flows that enhance the treasury reserves.
- As these reserves grow, they drive higher yields for onchain savers, increasing the demand for the synthetic dollar and continuously fueling the cycle.

2. Underlying Collateral Assets and Digital Asset Treasuries (DATs)

The primary collateral backing the Apyx protocol consists of preferred stock issued by Digital Asset Treasury (DAT) companies.

- **STRC (Strategy Inc.):** This is a preferred stock issued by Strategy (formerly MicroStrategy) that is listed on the Nasdaq. It features a \$100 par value and pays a

variable 12% annualized dividend. Strategy actively defends the par value, having recently bought back ~\$190M of STRC and built its USD reserves to \$4.65B, providing 31.7 months of dividend coverage.

- **SATA (Strive Inc.):** This instrument is a Variable Rate Series A Perpetual Preferred Stock with a stated amount of \$100 per share. It pays a 13% coupon (variable) as dividends accrue on the stated amount. The capital raised from SATA issuances is specifically utilized to acquire bitcoin.

3. Apyx's Architectural Engineering and Dual-Token System

The protocol utilizes a two-token architecture to securely separate liquid cash operations from yield accrual mechanisms.

apxUSD (The Synthetic Dollar)

- It is an over-collateralized, dividend-backed synthetic dollar supported by a reserve of crypto-related dividend-bearing real-world assets.
- Unlike traditional fiat-backed stablecoins, apxUSD is not directly redeemable for its underlying preferred shares.
- All minting and redemption operations for apxUSD are settled strictly in USDC.

apyUSD (The Yield Token)

- This is the yield-bearing stablecoin (YBS) of the protocol, built using the permissionless ERC-4626 vault standard.
- The token utilizes a non-rebasing, accrual-based model, meaning token balances in user wallets remain constant.
- Instead of receiving more tokens, yield is generated through a gradually increasing exchange rate, allowing each apyUSD to be redeemed for more apxUSD over time.
- The exchange rate is dynamically calculated using the following model:

$$\text{Exchange Rate} = \frac{\text{Total Underlying Assets (apxUSD)}}{\text{Total apyUSD Supply}}$$

4. Liquidation and Redemption Pathways

Apyx has designed secure, staggered redemption paths to prevent sudden liquidity shocks to the treasury.

- **Standard Redemption:** This asynchronous model streams the protocol's yield to apyUSD holders over a 20-day period.
- **Flexible Redemption:** Users initiating this path receive an onchain Unlock Receipt NFT representing their pending claim. The assets become claimable after 3 days, subject to an early redemption fee that declines linearly from 3.5% down to 0.1%.

5. Yield Comparison and DeFi Composability

Apyx offers a multi-layered yield ecosystem that consistently outperforms traditional stablecoins by providing double-digit returns.

- **Native Yield:** Holding apyUSD natively on Ethereum, BNB, SOL, or Base delivers a 21.15% native APR.
- **Leveraged Looping Strategies:** Users can loop their positions to maximize returns, with the Arkonix APYX Yield Vault currently offering 45.27% APR.
- **Morpho Integrations:** Manually looping PT-apyUSD / apxUSD on Morpho at 3.50x to 4.47x leverage achieves a 32.50% to 36.52% APR.
- **Fixed Yield via Pendle:** Users preferring predictable income can utilize Pendle markets for a 17.87% fixed yield on PT-apyUSD.

6. Cross-Chain Infrastructure and Solana Expansion

- **Solana Launch:** Apyx has launched on Solana to bring dividend-backed yield to the ecosystem, backed by a Superteam Earn bounty to educate users.
- **Institutional Alignment:** The expansion is supported by the team behind DeFi Development Corp. (Nasdaq: DFDV), which is the first SOL DAT.

- **Cross-Chain Capability:** Apyx leverages Chainlink CCIP and Data Feeds to safely expand apxUSD and apyUSD beyond Ethereum to networks like Base and BNB Chain.

7. APYX Tokenomics and the Pips (Points) Program

APYX Staking Vaults

- **xAPYX:** Users stake APYX to earn protocol revenue denominated in APYX. The vault buys APYX on the market and feeds it back to the vault, compounding the user's exposure.
- **yAPYX:** Users stake APYX and receive dollar-denominated dividends. Protocol revenue accrues in apxUSD and is distributed to yAPYX holders, making it ideal for those seeking predictable cash flows.

Apyx Pips Season 2

- **Reward Pool:** Season 2 features a larger reward pool, allocating 6,000,000 APYX, which represents 6% of the total token supply.
- **Loyalty Boost:** Wallets active during Season 1 automatically receive a 1.2x Loyalty Multiplier on all eligible Season 2 activity.
- **Multipliers:** High commitment yields extreme multipliers; committing apxUSD grants a 196x multiplier, while holding Pendle apxUSD YT yields 128x.

8. Transparency and Institutional Proof of Reserves

- **Accountable Dashboard:** A live, public dashboard powered by Accountable connects directly to the accounts where the protocol holds assets to continuously verify balances.
- **Third-Party Audits:** To guarantee institutional-grade accountability, the reserves undergo rigorous monthly attestations by the independent accounting firm Wolf & Company.

9. Operational Guide: How to Access Digital Credit

- **Step 1 (Acquire apxUSD):** General users can acquire apxUSD by swapping USDC on secondary markets through the Apyx Swap tab, executing trades via Curve.
- **Step 2 (Lock for Yield):** Using the Apyx Earn tab, users can lock their apxUSD into the vault to receive apyUSD shares and begin earning protocol-generated rewards.
- **Step 3 (Commit for Pips):** Users can head to the Rewards tab to commit apxUSD or Curve LP tokens to earn higher points multipliers ahead of the upcoming airdrop.

Here is the English translation of the text, with citations formatted according to your guidelines:

Digital Credit and the Apyx Financial Protocol Engineering

1. Comparing apyUSD Yields with Other DeFi Yields The Apyx protocol provides a solution to the missing yield dilemma in traditional stablecoins, outperforming other decentralized finance (DeFi) options:

- Traditional stablecoins (like USDT and USDC) suffer from generating little to zero automatic yield for users, exposing them to a loss of purchasing power due to inflation.
- In contrast, apyUSD offers exceptional double-digit yields to its holders.
- The direct native APR for holding apyUSD across networks like Ethereum, BNB, Solana, and Base can reach up to 21.15%.
- The platform offers yield-looping strategies, where the annual yield can reach 45.27% when using managed vaults like the "Arkonix APYX Yield Vault".
- The project also enables fixed and variable yields via the Pendle platform, reaching up to 17.87% as a fixed yield for Principal Tokens (PT), and up to 16.82% for Liquidity Pools (LP).

2. What is Digital Credit? Digital Credit represents an innovative asset class that connects traditional financial markets (TradFi) and decentralized finance (DeFi):

- Traditional credit markets are valued at trillions of dollars, but they have historically remained isolated and largely inaccessible within the blockchain ecosystem.
- Digital Credit is a new asset class that brings dividend-backed yields from publicly traded preferred equity into the cryptocurrency world.
- This class relies on Digital Asset Treasury (DAT) companies that issue variable-rate perpetual preferred stock to finance their balance sheets.

3. Why do STRC and SATA Matter? These stocks form the backbone of the collateral basket backing the Apyx protocol:

- **STRC stock:** It is a perpetual preferred stock issued by Strategy with a par value of \$100.
- STRC pays an annual dividend of 11.25%, which is distributed in cash on a monthly basis.
- STRC is classified as a short-term high-yield credit or a Bitcoin-backed money market.
- **SATA stock:** It represents the perpetual preferred stock (Series A) for Strive, with a stated par value of \$100.
- SATA pays a high annual dividend yield of 12.25%, which is also distributed monthly.
- The dividend rate for these shares is adjusted monthly to encourage trading near their par value and to limit volatility.

4. How Does the Apyx Project Work? The project relies on a two-token financial architecture to ensure stability and yield flow:

- The protocol issues apxUSD, which is an over-collateralized "synthetic dollar" backed by dividend-bearing real-world assets, rather than a traditional fiat-backed stablecoin.

- Users deposit USDC, and the offchain treasury uses these funds to purchase a collateral basket that includes preferred shares (STRC and SATA) and short-term treasuries.
- Users receive apyUSD (the yield-bearing version) when they lock their apxUSD in a permissionless vault built on the ERC-4626 standard.
- Yields generated from dividends accrue automatically through a gradually increasing exchange rate of apyUSD against apxUSD over time, without the need for a rebasing mechanism (non-rebasing).
- Transparency is ensured via an Accountable Dashboard that provides direct Proof-of-Reserves, alongside monthly attestation reports from the third-party firm Wolf & Company.

5. Why Did the Apyx Project Choose to Expand to the Solana Network? This

expansion aimed to achieve specific technical and economic goals:

- The Solana network aims to provide ultra-fast transaction processing with near-zero gas fees compared to the Ethereum network.
- This reduction in frictionless gas fees facilitates access for a larger segment of users to digital credit yields with high efficiency.
- The protocol enjoys strategic support from the team behind DeFi Development Corp., listed on Nasdaq, which is the first Digital Asset Treasury dedicated to the Solana network.
- This expansion creates a multi-chain liquidity environment, complementing the project's presence on the Ethereum, BNB Chain, and Base networks.

6. How Can Users Access Digital Credit Through Apyx? The user experience is simple and decentralized through clear steps:

- **Step One:** Users can swap USDC to acquire apxUSD (the synthetic dollar) on decentralized secondary markets (such as Curve) using the Apyx Swap interface.

- **Step Two:** The user locks apxUSD in the decentralized vault to receive shares represented by the yield-bearing apyUSD token.
- **Step Three:** Yields accrue automatically without manual intervention through the increasing exchange rate of apyUSD over time.
- **Additional Options:** Users can commit their assets, such as apxUSD or liquidity provider tokens (Curve LP), to earn exceptional points multipliers within the Apyx Pips program.